Stock Market Analysis in SQL: Insights Report

Six NSE stocks (Bajaj Auto, Eicher Motors, Hero Motocorp, Infosys, TCS, TVS Motors) | 1 Jan 2015 - 31 Jul 2018 | 889 trading days each
Prachi Sable | Code: stock_market_sql_analysis_FULL.sql | Live app: Streamlit dashboard + SQL playground

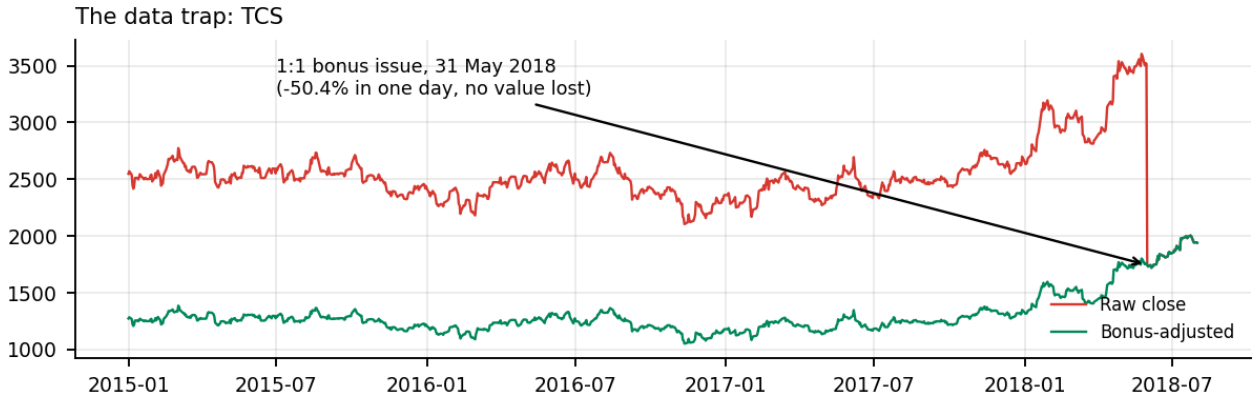
1. Business problem and objectives

A retail investor wants rule-based Buy/Sell signals for six NSE stocks that they can trust. The signal tested is the golden cross: a Buy when the 20-day average close crosses above the 50-day average, a Sell when it crosses below. The risk is bad data: raw prices contain jumps that are not real losses, which can distort returns and create false signals.

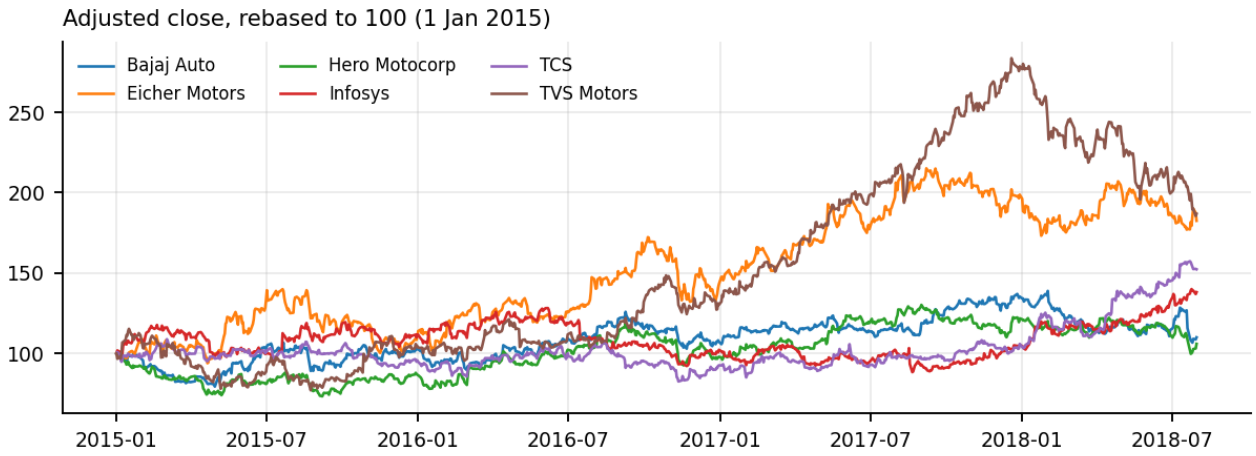
- **O1** Clean and validate the data and justify each decision.
- **O2** Profile return, risk, correlation and seasonality (EDA).
- **O3** Apply advanced SQL: window functions, CTEs, views, RSI, drawdown, rolling regression.
- **O4** Compare signal models on equal terms and test whether any beats buy-and-hold after costs.

2. Data preprocessing: what was found and decided

Check (SQL part 0)	Finding	Decision and reason
Missing values	1 row per stock has NULL deliverable_qty and pct_deli_qty (one date per stock; no price columns missing)	Kept, not imputed. It is a reporting gap, and delivery volume cannot be invented. Dropping the row would break the consecutive-day moving-average windows. Delivery analysis filters NULLs.
Integrity	0 duplicate dates, 0 rows with high below low, 0 closes outside the day range, 0 non-positive prices	No cleaning needed.
Calendar gaps	Largest gap is 5 days (weekends and holidays)	Rows treated as consecutive trading days, so a 20-row window is a 20-trading-day window.
Outliers (|z| > 5)	8 moves flagged. Two are about -50%: TCS on 2018-05-31 (-50.4%) and Infosys on 2015-06-15 (-49.9%)	Both are 1:1 bonus issues (price halves, holders lose nothing). Prices before each date divided by 2 (view adj_prices). Daily volatility drops from 2.17% to 1.36% (TCS) and 2.25% to 1.50% (Infosys).
Genuine shocks	Several -9% days, e.g. Bajaj and Eicher on 2015-08-24	Kept: they are real market risk, not data errors.



3. Exploratory analysis



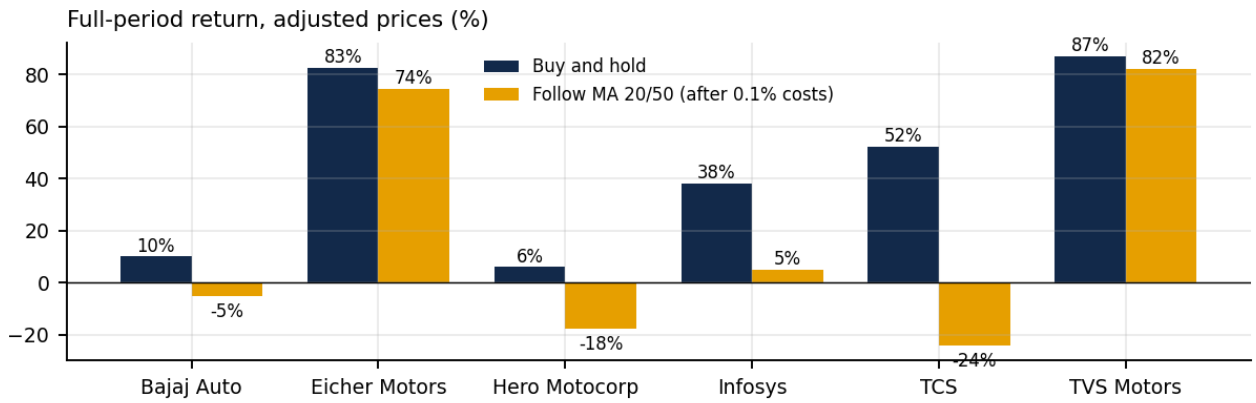
Stock	Total return	CAGR	Annual volatility	Return per unit risk	Max drawdown	2015	2016	2017	2018 (7 mo)
Eicher Motors	+82.6%	18.3%	29.1%	0.63	-28.3%	10.6	25.6	34.6	-7.0
TVS Motors	+86.9%	19.1%	30.8%	0.62	-34.6%	4.6	25.6	110.7	-32.5
TCS (adj.)	+52.4%	12.5%	21.6%	0.58	-24.1%	-4.3	-2.2	14.5	46.3
Infosys (adj.)	+38.2%	9.4%	23.8%	0.40	-31.1%	11.8	-8.4	3.8	31.9
Bajaj Auto	+10.0%	2.7%	23.3%	0.12	-23.2%	3.2	4.5	28.0	-18.0
Hero Motocorp	+6.0%	1.6%	22.3%	0.07	-26.9%	-13.3	13.3	25.0	-12.6

Yearly columns are % returns (last close of the year vs first close). 2018 covers January to July only, so it should not be called the "best" or "worst" year.

- **Diversification:** the highest return correlations are Bajaj-Hero (0.52) and Infosys-TCS (0.41); autos and IT are weakly linked (about 0.1-0.2), so mixing them lowers risk.
- **RSI test:** after oversold readings (RSI below 30) the next 5-day return was above each stock's average for all six stocks (for example Eicher +2.65% vs +0.43% on all days).
- **Volatility regime:** on 31 Jul 2018 Bajaj's 20-day volatility (2.78%) sat at the 99th percentile of its own history.

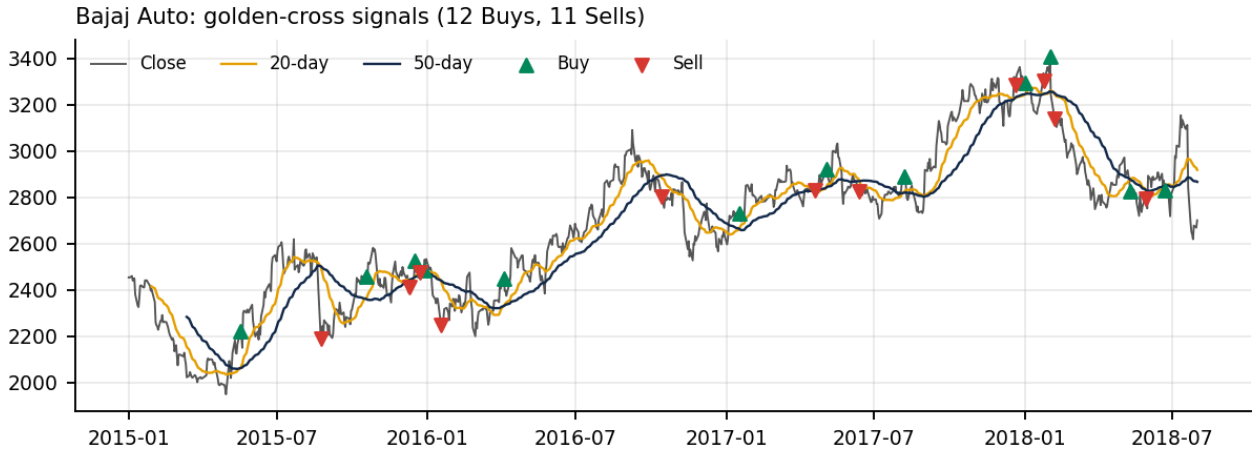
4. Findings per stock: claim, evidence, caveat

Stock	Claim	Evidence (SQL)	Caveat
Bajaj Auto	Flat stock; the signal added nothing.	Task 11/1.3: +10.0% over the period. Signals on adjusted prices: 12 Buys, 11 Sells; latest signal Buy on 2018-06-21. Following signals returned -5.0% vs +10.0% buy-and-hold (0.1% cost each side).	12 round trips, 5 lasted 30 days or less (avg -4.3%, 4 lost); win rate 33%. Latest signal conflicts with the long-run trend (price below its 200-day average, -5.1% over 12 months).
Eicher Motors	Strong winner; signals captured most of the rise.	Task 11/1.3: +82.6% over the period. Signals on adjusted prices: 6 Buys, 7 Sells; latest signal Sell on 2018-06-06. Following signals returned +74.4% vs +82.6% buy-and-hold (0.1% cost each side).	6 round trips, 0 lasted 30 days or less; win rate 100%. Latest signal agrees with the long-run trend (price below its 200-day average, -4.3% over 12 months).
Hero Motocorp	Weakest stock; signals lost money in a flat market.	Task 11/1.3: +6.0% over the period. Signals on adjusted prices: 9 Buys, 9 Sells; latest signal Sell on 2018-05-22. Following signals returned -17.8% vs +6.0% buy-and-hold (0.1% cost each side).	9 round trips, 1 lasted 30 days or less (avg -6.7%, 1 lost); win rate 33%. Latest signal agrees with the long-run trend (price below its 200-day average, -11.2% over 12 months).
Infosys	Real gain of +38% once adjusted; signals missed most of it.	Task 11/1.3: +38.2% over the period. Signals on adjusted prices: 10 Buys, 10 Sells; latest signal Buy on 2018-05-07. Following signals returned +5.0% vs +38.2% buy-and-hold (0.1% cost each side).	10 round trips, 2 lasted 30 days or less (avg -3.5%, 2 lost); win rate 40%. Latest signal agrees with the long-run trend (price above its 200-day average, +40.4% over 12 months).
TCS	Real gain of +52% once adjusted; signals lagged badly.	Task 11/1.3: +52.4% over the period. Signals on adjusted prices: 12 Buys, 12 Sells; latest signal Buy on 2018-04-20. Following signals returned -24.2% vs +52.4% buy-and-hold (0.1% cost each side).	12 round trips, 1 lasted 30 days or less (avg -4.6%, 1 lost); win rate 17%. Latest signal agrees with the long-run trend (price above its 200-day average, +56.3% over 12 months).
TVS Motors	Best performer; signals captured most of it with large drawdowns.	Task 11/1.3: +86.9% over the period. Signals on adjusted prices: 8 Buys, 8 Sells; latest signal Sell on 2018-05-17. Following signals returned +82.1% vs +86.9% buy-and-hold (0.1% cost each side).	8 round trips, 2 lasted 30 days or less (avg -8.5%, 2 lost); win rate 38%. Latest signal agrees with the long-run trend (price below its 200-day average, -9.8% over 12 months).



Flip-flop stretches (whipsaws) were costly: Bajaj's short trades lost 4.3% on average, TVS's 8.5%, Hero's 6.7%. Short losing round trips like these are the main cost of the strategy in sideways markets.

5. Golden-cross signals



Across all six stocks there are 56 Buys and 57 Sells (matches the guide). Buy and Sell counts differ by at most one per stock, because after a Buy the next cross must be a Sell. The first Bajaj Buy is on 2015-05-18 and the first Sell on 2015-08-24.

Effect of the bonus-issue fix (stretch task): on raw prices TCS shows 12 Buys / 13 Sells with a last signal of **Sell on 2018-06-05**; on adjusted prices it is 12 / 12 with a last signal of **Buy on 2018-04-20**. Infosys moves from 9 / 9 to 10 / 10. Signals that exist only because of the price cliff: TCS 1, Infosys 1; real signals the cliff hid: TCS 0, Infosys 3.

6. Systematic model comparison

Each model decides at the close whether to hold the stock tomorrow. All are scored on the same days (after day 200) with a 0.1% cost on every position change, averaged over six stocks.

Model	Avg return	Direction accuracy	Avg trades	Stocks beating buy-and-hold	Rank
Buy and hold	46.0%	49.7%	0	-	1
MA 50/200	40.1%	49.5%	4.3	2	2
MA 20/50 (assignment signal)	23.1%	48.6%	15.0	1	3
Rolling 20-day regression trend	17.3%	48.3%	33.7	0	4
MA 5/20	14.5%	48.0%	41.7	0	5
MA 10/30	12.1%	47.7%	28.0	0	6
1-day momentum	1.0%	49.0%	351.0	0	7

Reading: no model beats buy-and-hold on average. Slower signals generally do better, because costs and whipsaws eat the faster ones. Direction accuracy of all models sits near 50%, which is about chance. The "models" are rule-based or a closed-form regression, not trained machine-learning models.

7. Answers to the guide's method questions

- **Lag of a moving-average cross:** a 20/50 cross needs 50 trading days of history before the first signal (earliest here: 2015-03-30), and it only fires after the averages have already turned. By then part of the move has happened, and a reversal can follow within weeks, which helps explain why short trades lost money.
- **What changes after the bonus-issue fix:** TCS and Infosys go from "losers" (-23.8%, -30.9%) to winners (+52.4%, +38.2%); their daily volatility falls by about a third; TCS's latest signal flips from Sell to Buy; and the ranking by return becomes TVS, Eicher, TCS, Infosys, Bajaj, Hero. Conclusions that rely on raw TCS or Infosys prices should be re-checked; compare these figures with the course deck before submitting.
- **What is missing from the data:** dividends, corporate-action confirmation from company filings, news, taxes and real brokerage and slippage (a flat 0.1% is assumed), the broader market index for comparison, and fundamentals. The window is 3.6 years and mostly a rising market.

8. Conclusion and recommendations

- Always adjust for corporate actions before computing returns or signals; here it reversed the story for two of six stocks.
- A standalone 20/50 golden cross did not beat buy-and-hold in this period. Use it as a trend filter alongside risk limits, not as a stand-alone trading rule.
- Mix autos and IT for diversification, and size positions by drawdown tolerance: TVS fell 34.6% from its peak even though it was the best stock.

Educational project, not investment advice. Past performance does not predict future returns. Results are specific to six stocks and the period shown.